

CAPSTONE PROJECT

AgriSense.AI

Solving South Africa's water crisis in agriculture, one photo at a time.

AI Diagnosis · Water Solutions · Urban + Rural Farmers

Built with TanStack Start · Lovable Cloud · Google Gemini 2.5 Flash

South Africa is running dry.



60%

of SA's freshwater
goes to agriculture

98%

of available water
is already allocated

1 in 3

municipalities face
weekly water cuts

Two farmers. One platform.

Urban Smallholder

Soweto, Khayelitsha, Mamelodi

- Backyard & community gardens
- Tomatoes, spinach, peppers, herbs
- Limited water, no extension officer
- Smartphone-first, low data budget

Rural Commercial Farmer

Free State, Limpopo, Eastern Cape

- Maize, livestock, citrus, vines
- Boreholes, dams, drip & flood irrigation
- Load-shedding hits pumps & cold-chain
- Needs fast diagnosis at scale

AgriSense.AI in one breath.

A web app that lets any South African farmer snap a photo of a sick plant, thin cow, dry soil or leaking pipe — and get an instant, water-first diagnosis with a 3-step action plan.



Diagnose

Image AI for crops, livestock, soil & water.



AI Advisor

Chat for SA-specific agronomy questions.



Water-First

Every answer prioritises saving water.

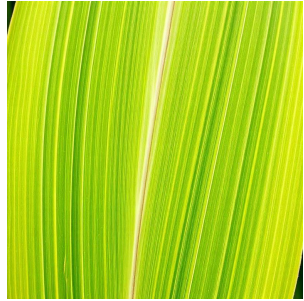


Connected

Links farmers to finance, transport & markets.

One-tap sample diagnosis.

Six clickable demo images — judges & first-time farmers can experience the AI without uploading a single file.



 Sick maize leaf



 Underweight cattle



 Wilted vegetables



 Leaking irrigation

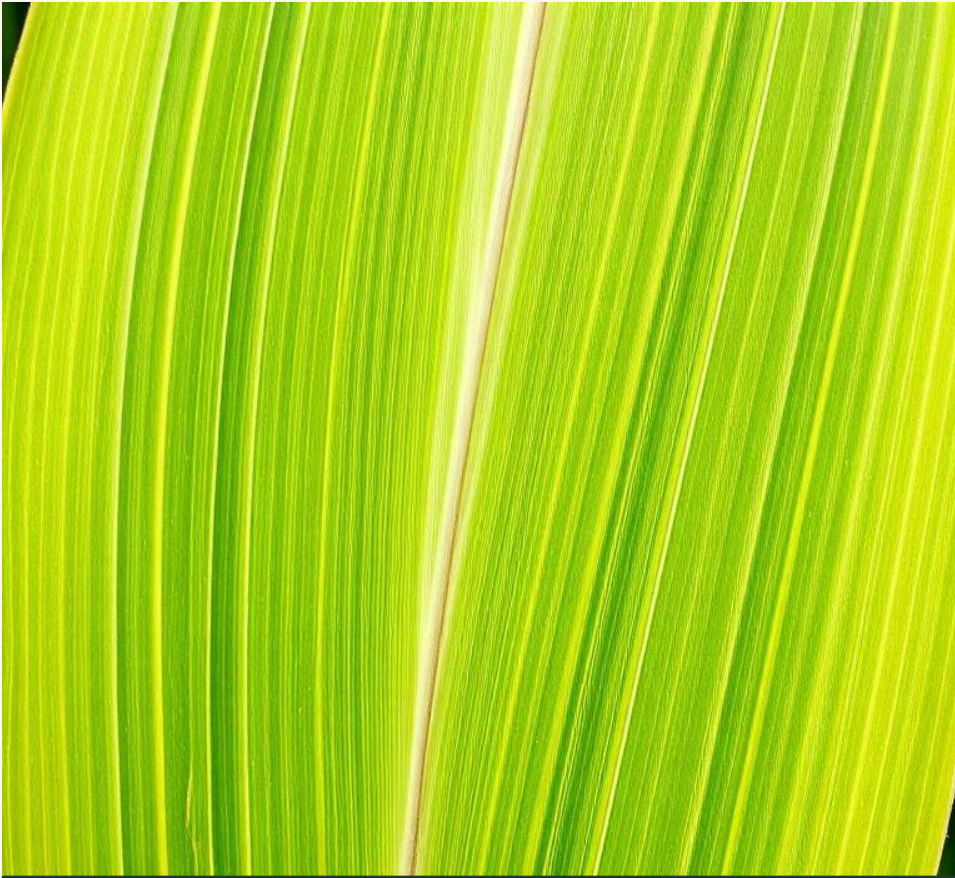


 Cracked dry soil



 Tomato blight

Maize leaf — nitrogen + water stress



 Uploaded by farmer · Diagnosed by Gemini 2.5 Flash

AI Diagnosis

Yellowing between the veins on lower leaves with a V-shape from the tip suggests nitrogen deficiency, often worsened by water stress. Soil is likely dry to root depth.

Immediate (today)

Deep-water at root zone in early morning. Mulch with dry grass to lock moisture.

This week

Side-dress 30 g LAN per plant after rain. Scout daily for fall armyworm.

Long-term

Rotate with cowpea/legume next season; install drip on this block.



Underweight Nguni — BCS 2/5



📷 Uploaded by farmer · Diagnosed by Gemini 2.5 Flash

AI Diagnosis

Visible ribs, sunken flanks and dull coat. Body Condition Score around 2 out of 5. Likely protein and energy shortfall on dry winter veld.

Immediate (today)

Move to shaded camp with clean trough water within 200 m of grazing.

This week

Supplement 500 g/day protein lick (urea-based). Deworm and check ticks.

Long-term

Reduce stocking rate; oversow with Eragrostis for winter standing hay.

Cracked dry soil — severe moisture loss



Uploaded by farmer · Diagnosed by Gemini 2.5 Flash

AI Diagnosis

Wide polygonal cracks indicate clay soil that has lost most of its water. Capillary rise is exposing the root zone — seedlings will scorch.

Immediate (today)

Cover bare soil with 5 cm mulch (straw, cardboard, leaves) before noon.

This week

Build shallow swales across slope; harvest greywater into a 200 L drum.

Long-term

Switch to drip + plant cover crops (sun hemp) to rebuild organic matter.

Chat: a farmer in Limpopo at 06:12.

Sawubona. My borehole is giving brown water and my tomatoes wilt by 10am. What do I do?

Sanibonani. Brown borehole water = iron/sediment after a heavy draw or low water table. Don't irrigate tomatoes with it directly.

- Today: Pump 10 min into a settling drum, let it stand 2 hrs, water from the top.
- This week: Fit a 50-micron sediment filter at the outlet (R350 at most co-ops).
- Long-term: Test borehole yield + harvest rainwater (~2 000 L per 20 mm rain on 100 m²).

Eish, and the wilting?

10am wilt with cool nights = water stress + heat, not disease yet. Mulch with 5 cm dry grass, add 40% shade-cloth, water at 5am only. If leaves curl upward tomorrow, send a photo.

Three apps already in the field.

Plantix

Crop disease photo ID

Strength

Huge image library

Gap

No SA water focus, no livestock

Aerobotics

Drone + AI for orchards

Strength

High-precision, used by SA citrus

Gap

Enterprise-only, expensive

Khula!

Marketplace for SA smallholders

Strength

Connects farmers to buyers

Gap

No diagnosis or agronomic advice

Agriculture touches every part of our lives.

Manufacturing

Agro-processing, packaging, agri-tech

Water

60% of national water → smarter irrigation

Transport

Cold-chain & rural roads to market

Agriculture

Finance

Crop loans, insurance, micro-credit

Energy

Pumps & cold rooms hit by load-shedding

Health

Food security & nutrition outcomes

How it works under the hood.

1. Client (TanStack Start + React 19)

Routes: /, /diagnose, /advisor, /insights · Tailwind v4 · sonner · react-markdown

2. Edge (Lovable Cloud)

Supabase Edge Function `agri-ai` · CORS · request validation · error mapping (402/429)

3. AI (Lovable AI Gateway)

google/gemini-2.5-flash · multimodal (text + image_url) · system prompt = SA water-first agronomist

4. Storage (future)

Supabase Postgres + Storage for diagnosis history, farm profiles, RLS by user

Photo → /diagnose → fileToBase64 → supabase.functions.invoke('agri-ai') → Gemini → markdown plan → ReactMarkdown render

How AgriSense actually saves water.



Drip + mulch defaults

Every irrigation answer leads with drip and 5cm mulch — cuts use 30–60%.



Rainwater harvesting

Calculates litres-per-roof and recommends JoJo sizes by district rainfall.



Greywater reuse

Safe-use guidance for kitchen/laundry water on non-edible parts.



Time-of-day scheduling

Suggests pre-dawn watering windows aligned to load-shedding stages.



Leak detection

Photo-diagnoses leaking pipes, valves and float-valves with fix-it steps.



Drought-tolerant choices

Recommends sorghum, cowpea, amaranth swaps for high-water crops.

From hackathon to harvest.

1

Now

Web app live · 6 sample diagnoses · AI advisor chat

2

Next 30 days

isiZulu + Sesotho · offline PWA · WhatsApp bot

3

6 months

Soil moisture sensor integration · weather alerts · co-op partnerships

4

12 months

On-device vision for zero-data farms · finance & insurance hooks

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